

Ling Tao

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Montreal, Quebec, Canada

Full Stack Developer

About Me

I am a **Full-Stack Developer** with extensive experience designing and optimizing **scalable, high-performance web applications**. Specialized in building front-end and back-end API systems and event-driven architectures using **Node.js, TypeScript, React, Next.js, Python, PostgreSQL, MongoDB, AWS**, and **microservices**. Skilled in **3D visualization and simulation** with **Three.js, Shaders, React Three Fiber/Drei, WebGL, and WebGPU**. Proficient in leveraging a wide range of **AI models and tools** to accelerate development and enhance productivity. Adaptable, detail-oriented, and passionate about applying **AI and advanced web technologies** to deliver innovative, high-impact digital solutions.

Work Experience

Full Stack Developer, NEMCCO Consulting, Québec, Canada 01/2023 - 07/2025

- **Led full-stack architecture design:** Built a 3D Underground Mining Data Management platform, improving data visualization and collaboration across multiple departments.
- **3D Model Management:** Built bulk 3D model upload & optimization system supporting DXF, GLB, and PLY formats, improving web performance and loading speed by ~45%.
- **Access Control System:** Designed and implemented a policy-based user permission system, dynamically rendering interfaces based on user privileges to enhance collaboration across departments, sites, and organizations.
- **Reporting & Activity Tracking System:** Designed a comprehensive reporting and planning tool integrated with 3D model positions. Enabled team managers to plan activities, assign tasks, and send email notifications. Allowed team members to input data, upload on-site photos, and receive approvals.
- **Custom Form Builder:** Designed a flexible form data structure enabling users to create custom forms and calculators.
- **Admin & File Management System:** Created an admin dashboard to manage users, files, and data, including a folder tree component for fast file organization and access.
- **3D Simulation:** Designed a seismic impact simulation model to calculate reinforcement requirements based on user input, significantly reducing operational costs for users.
- **3D Animations:** Implemented 3D animation and time-sequenced visualization features to simulate tunnel development, boosting project clarity for engineering teams..
- **3D Measurement Tools:** Built a 3D ruler feature for measuring distances and positions across models, boosting accuracy and efficiency for engineering teams.
- **3D Data Visualization:** Built features to visualize geotechnical data on 3D models, helping users identify issues and discover patterns.
- **Power BI Integration:** Developed Activity Data APIs to synchronize analytics with Microsoft Power BI, enabling dynamic reporting tied to model coordinates and real-time operational insights.

- **Data Import/Export:** Implemented bulk data handling with support for Excel, CSV, and JSON formats, boosting data transfer efficiency
- **Offline Web Application:** Enabled offline capabilities for model viewing, data entry, and form submission, ensuring uninterrupted field operations even without network access.
- **Server Maintenance:** Managed and monitored Linux-based remote servers (Ubuntu) using PuTTY and SSH to ensure reliable daily ETL processes.
- **Main Technologies Used:** Node.js, TypeScript, React, JavaScript, Python, Three.js, GLSL Shaders, React Three Fiber/Drei, WebGPU, WebGL, D3.js, MongoDB, Blender, Power BI

Researcher, Polytechnique Montréal , Québec, Canada **05/2014 - 04/2019**

- Conducted R&D on lithium-ion battery materials; published in The Canadian Journal of Chemical Engineering.

Engineer & Researcher, Institute of Process Engineering, China **07/2007 - 06/2013**

- Designed and optimized pilot-scale reactors and automation control systems.

Technical Skills

Languages & Frameworks: **TypeScript, Node.js, React, JavaScript, NextJS, Python**
3D & Visualization: **Three.js, Shaders, React Three Fiber/Drei, WebGPU, WebGL, D3.js, Blender**
Databases: **PostgreSQL, MongoDB, MySQL**
Cloud & Tools: **AWS, Ubuntu, CI/CD, REST APIs, JWT**
UI & Styling: **HTML, CSS, Sass, Tailwind, Material UI, Bootstrap**
Data & Analytics: **PowerBI, Looker**
AI Tools: **ChatGPT, Claude, GitHub Copilot, Cursor, Visual Studio Code**

Education	AEC in Full Stack Development John Abbott College	01/2022 – 04/2023
	M.Sc. in Chemical Engineering Institute of Process Engineering	09/2004 – 07/2007
	B.Sc. in Chemical Engineering University of Mining and Technology	09/2000 - 07/2004

Languages

English, French, Mandarin

